BRAC University

Department of Mathematics and Natural Sciences

Lecture on

Real Analysis II (MAT321)

Separable Spaces

Dense Set and Perfect Set

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Conducted by

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Dense Sets

Dense Set

A subset A of a metric space (X, d) is said to be *dense* (or *everywhere dense*) in X if the closure of A is X, that is,

$$\overline{A} = X.$$

Nowhere Dense Sets

Nowhere Dense Set

Let A be a subset of a metric space (X, d). The set A is said to be *nowhere dense* in X if the interior of its closure is empty, that is,

$$\operatorname{int}(\overline{A}) = \emptyset.$$

Basic Properties of Nowhere Dense Sets

Show that:

1. every finite subset of X is nowhere dense;
2. every subset of a nowhere dense set is nowhere dense;
3. A is nowhere dense in X if and only if $\overline{A}$ is nowhere dense in X.

Somewhere Dense Sets and Dense-in-itself Sets

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Somewhere Dense Set

A subset A of X is said to be *somewhere dense* in X if it is not nowhere dense. Equivalently, A is somewhere dense in X if and only if $\overline{A}$ contains a non-empty open sphere.

Dense-in-itself Set

A subset A of a metric space (X, d) is said to be *dense-in-itself* if every point of A is a limit point of A, that is,

$$A \subseteq A'.$$

Perfect Sets

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Perfect Set

A subset A of a metric space (X, d) is said to be *perfect* if it is closed and dense-in-itself, that is,

$$A = A'.$$

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Examples of Perfect Sets

Show that every closed interval, the empty set $\emptyset$, and the whole space X are perfect sets.

Cantor Set is Perfect

Show that the Cantor set is a perfect set.

Separable Metric Space

Separable Metric Space

A metric space (X, d) is said to be *separable* if there exists a countable subset of X which is dense in X.

Euclidean Space is Separable

Show that the Euclidean space $\mathbb{R}^n$ is separable.

ℓ^p is Separable

Show that the metric space ℓ^p of all sequences (x_n) of real numbers such that $\sum_{n=1}^{\infty} |x_n|^p < \infty$ with the metric

$$d_p((x_n),(y_n)) = \left(\sum_{n=1}^{\infty} |x_n - y_n|^p\right)^{1/p}$$

is separable for every $1 \leq p < \infty$.

ℓ^∞ is Not Separable

Prove that the metric space ℓ^∞ of all bounded sequences with the supremum metric is not separable.

Thank You!

We'd love your questions and feedback.

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(Lectures, walkthroughs, and course updates)

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References

[1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.

[2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.